

Sample Exam 2026

The exam has two parts.

Part 1 is to be completed with pen and paper.

Part 2 should be completed on your assigned computer via Blackboard.

Part 1: Question 1 (Pen & Paper)

(50 marks)

A university wants to design a database to manage information about its academic departments, lecturers, modules, and students. The requirements are as follows: each department has a department name, and an office location, each lecturer has a name, and email, and belongs to exactly one department. Furthermore, each module has a unique module code, a title, and a number of credits to be awarded for that module once completed. Each module is coordinated by exactly one lecturer, but a lecturer may coordinate many modules, a module is offered by one department, and each department may offer many modules. Each student has a unique student id number, a name, a date of birth, and an email address. A student can enrol in many modules, and a module can have many students. Additionally, for each enrolment, the university must store the enrolment date and the student's grade. A lecturer may act as an academic advisor for many students, but each student has at most one academic advisor. Some lecturers are also Heads of Department. Each department has exactly one Head of Department, and a lecturer can head at most one department.

(a) Draw an ER diagram that includes: all entities, primary keys, attributes, multivalued attributes, weak entities (if needed), specialization / generalization, relationship attributes, and cardinality and participation constraints. (30 marks)

(b) Map the ER diagram to relational tables, explaining each step as you go. (20 marks)

Part 2: Multiple Choice Questions (Computer & Blackboard)

(50 marks)

Each MCQ below is worth 5 marks.

MC Question 1:

A database administrator grants the following permissions:

GRANT SELECT, UPDATE ON Employees TO user1;

GRANT SELECT ON Employees TO user2;

Later, the administrator executes:

REVOKE SELECT ON Employees FROM user1;

Which of the following is TRUE?

- A. **user1** loses both **SELECT** and **UPDATE** privileges
- B. **user1** retains **UPDATE** but loses **SELECT**
- C. **user1** retains both privileges due to dependency
- D. **user1** cannot perform any operations until privileges are reassigned

MC Question 2:

Consider the following schedule involving two transactions:

- T1: **R(X)**, **W(X)**
- T2: **R(X)**, **W(X)**

Schedule:

R1(X), **R2(X)**, **W1(X)**, **W2(X)**

Which statement is correct?

- A. The schedule is conflict-serializable
- B. The schedule is view-serializable but not conflict-serializable
- C. The schedule is not serializable
- D. The schedule is strict

MC Question 3:

Which of the following scenarios BEST justifies using a document-oriented NoSQL database over a relational database?

- A. Strong ACID guarantees are required for financial transactions
- B. Data schema is fixed and highly normalized
- C. Frequent schema evolution with nested, semi-structured data
- D. Complex joins across multiple tables are required

MC Question 4:

Given the table:

`Orders(order_id, customer_id, amount)`

Which query returns customers whose total order amount exceeds 1000?

A.

```
SELECT customer_id
FROM Orders
WHERE SUM(amount) > 1000;
```

B.

```
SELECT customer_id
FROM Orders
GROUP BY customer_id
HAVING SUM(amount) > 1000;
```

C.

```
SELECT customer_id
FROM Orders
WHERE amount > 1000;
```

D.

```
SELECT DISTINCT customer_id
FROM Orders
HAVING SUM(amount) > 1000;
```

MC Question 5:

Under GDPR, which of the following is considered a *lawful basis* for processing personal data?

- A. Data is publicly available
- B. User consent has been explicitly obtained
- C. The company benefits financially
- D. Data is anonymized after processing

MC Question 6:

Consider the following SQL constraint:

```
ALTER TABLE Employees  
ADD CONSTRAINT chk_salary CHECK (salary > 0 AND salary < 100000);
```

Which statement is TRUE?

- A. The constraint is enforced only during insertion
- B. The constraint can be bypassed using transactions
- C. The constraint is enforced on both INSERT and UPDATE operations
- D. The constraint guarantees uniqueness of salary values

MC Question 7:

A company based in Ireland operates an e-commerce platform and processes customer data (names, emails, purchase history).

Scenario:

1. Users initially give consent for data processing during account creation
2. The company later decides to use the same data for targeted advertising and update consent with the users
3. A user requests erasure of their data
4. The company deletes the user's personal data from its main database but retains it in backups and analytics datasets

Which of the following is the MOST accurate assessment under GDPR?

- A. The company is compliant because initial consent was originally gotten and then updated to include advertising
- B. The company is non-compliant due to unlawful processing and incomplete erasure
- C. The company is compliant if data is only used internally for analytics
- D. The company is non-compliant because GDPR is a basic human right applicable worldwide

MC Question 8:

Consider two transactions operating under **strict two-phase locking (Strict 2PL)**:

- **T1:** R(X), W(X), R(Y), W(Y)
- **T2:** R(Y), W(Y), R(X), W(X)

Execution proceeds as follows:

1. T1 acquires a shared lock on X and reads it
2. T2 acquires a shared lock on Y and reads it
3. T1 upgrades to an exclusive lock on X and writes it
4. T2 upgrades to an exclusive lock on Y and writes it
5. T1 requests a lock on Y
6. T2 requests a lock on X

What will happen?

- A. Both transactions complete successfully without conflict
- B. A deadlock occurs between T1 and T2

- C. One transaction is rolled back due to dirty reads
- D. The schedule is serializable without blocking

MC Question 9:

Three transactions with timestamps (lower = older):

- **T1 (TS = 1)** → oldest
- **T2 (TS = 2)**
- **T3 (TS = 3)** → youngest

They request locks in the following order:

1. T2 acquires a lock on resource X
2. T3 acquires a lock on resource Y
3. T1 requests X
4. T2 requests Y
5. T3 requests X

Assume the system uses **Wait-Die**.

Which of the following correctly describes what happens?

- A. T1 aborts
- B. T3 aborts
- C. T1 waits, T3 aborts
- D. T2 aborts

MC Question 10:

The same scenario is repeated here from MC Question 9:

Three transactions with timestamps (lower = older):

- **T1 (TS = 1)** → oldest
- **T2 (TS = 2)**
- **T3 (TS = 3)** → youngest

They request locks in the following order:

1. T2 acquires a lock on resource X
2. T3 acquires a lock on resource Y
3. T1 requests X
4. T2 requests Y
5. T3 requests X

Assume the system uses **Wound-Wait**.

Which of the following correctly describes what happens?

- A. T1 waits
- B. T2 aborts
- C. T3 aborts
- D. T2 waits

End of Sample Exam